

Homework (Habanero)

- Q1.** What is the unit rate of $\frac{250 \text{ milligrams}}{5 \text{ milliliters}}$?
- (A) $50 \frac{\text{milligrams}}{\text{milliliter}}$
(B) $50 \frac{\text{milliliters}}{\text{milligram}}$
(C) $5 \frac{\text{milligrams}}{\text{milliliter}}$
(D) $5 \frac{\text{milliliters}}{\text{milligram}}$
(E) $25 \frac{\text{milligrams}}{\text{milliliter}}$
- Q2.** If a patient needs 3 units of insulin per 20 hours, what is the rate of insulin per hour?
- (A) $\frac{3}{20}$ units/hour
(B) $\frac{1}{6}$ units/hour
(C) $\frac{3}{4}$ units/hour
(D) $\frac{1}{5}$ units/hour
(E) 1 unit/hour
- Q3.** The volume V of a tumor is directly proportional to the cube of its radius r . If $V = 64$ cubic centimeters when $r = 4$ centimeters, what is the constant of proportionality k ?
- (A) 2
(B) 1
(C) 4
(D) $\frac{1}{2}$
(E) $\frac{1}{4}$
- Q4.** A lab technician can prepare 12 units of a solution in 4 minutes. At this rate, how many minutes will it take to prepare 30 units?
- (A) 5 minutes
(B) 10 minutes
(C) 8 minutes
(D) 12 minutes
(E) 15 minutes
- Q5.** The growth rate of a cell culture is inversely proportional to the concentration of a nutrient. If the culture grows at a rate of 2 cells per minute when the nutrient concentration is 6 milligrams per liter, what is the rate when the concentration is 12 milligrams per liter?
- (A) 1 cell/minute
(B) 2 cells/minute
(C) 4 cells/minute
(D) 6 cells/minute
(E) 8 cells/minute
- Q6.** A pharmaceutical company produces tablets with 200 milligrams of active ingredient per tablet. How many tablets are needed to obtain 4.8 grams of the active ingredient?
- (A) 12
(B) 16
(C) 20
(D) 24
(E) 8
- Q7.** A patient is given 250 milliliters of a solution that contains 12.5 milligrams of a medication per 100 milliliters. How much of the medication does the patient receive?
- (A) 25 milligrams
(B) 31.25 milligrams
(C) 12.5 milligrams
(D) 50 milligrams
(E) 62.5 milligrams
- Q8.** The volume V of a spherical cell is directly proportional to the cube of its radius r , given by $V = \frac{4}{3}\pi r^3$. If $V = 216\pi$ cubic micrometers, what is r ?
- (A) 6
(B) 3
(C) 4
(D) 2
(E) 1

Open Ended Questions (FRQ)

- Q9.** A bacterial culture grows at a rate that is inversely proportional to the concentration of a certain chemical. If the culture grows at 10 cells per hour when the concentration is 4 milligrams per liter, find the rate when the concentration is 2 milligrams per liter. Show all work and explain your solution.
- Q10.** A hospital is planning to purchase a new machine that can perform 20 tests per hour. The machine operates 8 hours per day and 5 days per week. If each test requires 2 milliliters of a certain chemical and the chemical costs

per 100 milliliters, what is the weekly cost of the chemical?

Chef's Tips

- Chef's Tips: Make sure to read each question carefully.
- Chef's Tips: Use the given information and apply the concepts of rates and variation.
- Chef's Tips: Show all work and explain your reasoning for free response questions.